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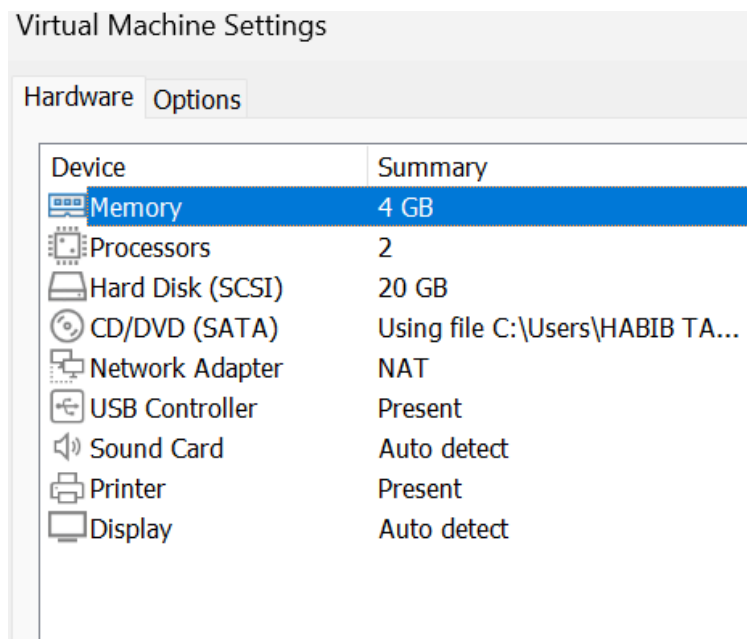
Section: 5-B

Reg no: 2023-BSE-042

Lab 4

Task 1

Confirm the VM resources that were allocated in Lab 1



The screenshot shows the 'Virtual Machine Settings' window with the 'Hardware' tab selected. It displays a list of devices and their configurations. The 'Memory' row is highlighted in blue.

Device	Summary
Memory	4 GB
Processors	2
Hard Disk (SCSI)	20 GB
CD/DVD (SATA)	Using file C:\Users\HABIB TA...
Network Adapter	NAT
USB Controller	Present
Sound Card	Auto detect
Printer	Present
Display	Auto detect

Task 2

Use a single preferred host-terminal method to connect to the VM. Do not switch between methods during the task

```

Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.58.139' (ED25519) to the list of known hosts.
manahil@192.168.58.139's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-85-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Oct 17 04:29:03 AM UTC 2025

System load:  0.0               Processes:    217
Usage of /:   49.0% of 9.75GB   Users logged in: 1
Memory usage: 8%               IPv4 address for ens33: 192.168.58.139
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

31 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

```

Whoami, pwd:

```

manahil@ubuntu:~$ whoami
manahil
manahil@ubuntu:~$ pwd
/home/manahil
manahil@ubuntu:~$

```

Task 3

1. List root directory contents:

```

manahil@ubuntu:~$ ls -la /
total 1994844
drwxr-xr-x 23 root root      4096 Sep 26 07:21 .
drwxr-xr-x 23 root root      4096 Sep 26 07:21 ..
lrwxrwxrwx  1 root root         7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x  2 root root      4096 Feb 26  2024 bin.usr-is-merged
drwxr-xr-x  4 root root      4096 Oct  2 08:25 boot
dr-xr-xr-x  2 root root      4096 Aug  5 23:53 cdrom
drwxr-xr-x 20 root root      4120 Oct 17 04:12 dev
drwxr-xr-x 108 root root     4096 Oct  2 08:24 etc
drwxr-xr-x  3 root root      4096 Sep 26 07:25 home
lrwxrwxrwx  1 root root         7 Apr 22  2024 lib -> usr/lib
lrwxrwxrwx  1 root root         9 Apr 22  2024 lib64 -> usr/lib64
drwxr-xr-x  2 root root      4096 Feb 26  2024 lib.usr-is-merged
drwx----- 2 root root     16384 Sep 26 12:18 lost+found
drwxr-xr-x  2 root root      4096 Aug  5 16:54 media
drwxr-xr-x  2 root root      4096 Aug  5 16:54 mnt
drwxr-xr-x  2 root root      4096 Aug  5 16:54 opt
dr-xr-xr-x 281 root root         0 Oct 17 04:11 proc
drwx----- 3 root root      4096 Aug  5 17:02 root
drwxr-xr-x 30 root root      880 Oct 17 04:29 run
lrwxrwxrwx  1 root root         8 Apr 22  2024/sbin -> usr/sbin
drwxr-xr-x  2 root root      4096 Dec 11  2024/sbin.usr-is-merged
drwxr-xr-x  2 root root      4096 Sep 26 07:25 snap
drwxr-xr-x  2 root root      4096 Aug  5 16:54 srv
-rw-----  1 root root 2042626048 Sep 26 07:21 swap.img
dr-xr-xr-x 13 root root         0 Oct 17 04:11 sys
drwxrwxrwt 15 root root      4096 Oct 17 04:25 tmp
drwxr-xr-x 12 root root      4096 Aug  5 16:54 usr
drwxr-xr-x 13 root root      4096 Sep 26 07:25 var
manahil@ubuntu:~$

```

2. View OS release information

```
manahil@ubuntu:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.3 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.3 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
manahil@ubuntu:~$
```

3. Inspect these directories (run each command and screenshot the output):

```
manahil@ubuntu:~$ ls -la /bin
lrwxrwxrwx 1 root root 7 Apr 22  2024 /bin -> usr/bin
manahil@ubuntu:~$ _
```

4.

```
manahil@ubuntu:~$ ls -la /bin
lrwxrwxrwx 1 root root 7 Apr 22  2024 /bin -> usr/bin
manahil@ubuntu:~$ ls -la /sbin
lrwxrwxrwx 1 root root 8 Apr 22  2024 /sbin -> usr/sbin
manahil@ubuntu:~$
```

```
manahil@ubuntu:~$ ls -la /opt
total 8
drwxr-xr-x  2 root root 4096 Aug  5 16:54 .
drwxr-xr-x 23 root root 4096 Sep 26 07:21 ..
manahil@ubuntu:~$
```

```
manahil@ubuntu:~$ ls -la /usr
total 104
drwxr-xr-x 12 root root 4096 Aug  5 16:54 .
drwxr-xr-x 23 root root 4096 Sep 26 07:21 ..
drwxr-xr-x  2 root root 36864 Oct  2 08:24 bin
drwxr-xr-x  2 root root 4096 Apr 22  2024 games
drwxr-xr-x 33 root root 12288 Sep 28 06:47 include
drwxr-xr-x 79 root root 4096 Oct  2 08:25 lib
drwxr-xr-x  2 root root 4096 Sep 28 06:47 lib64
drwxr-xr-x 11 root root 4096 Sep 26 07:21 libexec
drwxr-xr-x 10 root root 4096 Aug  5 16:54 local
drwxr-xr-x  2 root root 20480 Oct  2 08:23 sbin
drwxr-xr-x 124 root root 4096 Sep 26 07:22 share
drwxr-xr-x  6 root root 4096 Oct  2 08:25 src
manahil@ubuntu:~$
```

```

drwxr-xr-x 2 root root 4096 Aug 5 17:14 sensors.d
-rw-r--r-- 1 root root 12813 Mar 27 2021 services
drwxr-xr-x 2 root root 4096 Aug 5 17:02 sgml
-rw-r----- 1 root shadow 969 Sep 26 07:25 shadow
-rw-r----- 1 root shadow 969 Sep 26 07:25 shadow-
-rw-r--r-- 1 root root 148 Aug 5 17:14 shells
drwxr-xr-x 2 root root 4096 Aug 5 16:55 skel
drwxr-xr-x 6 root root 4096 Aug 5 17:14 sos
drwxr-xr-x 4 root root 4096 Sep 26 07:25 ssh
drwxr-xr-x 4 root root 4096 Oct 2 08:22 ssl
-rw-r--r-- 1 root root 21 Sep 26 07:25 subgid
-rw-r--r-- 1 root root 0 Aug 5 16:54 subgid-
-rw-r--r-- 1 root root 21 Sep 26 07:25 subuid
-rw-r--r-- 1 root root 0 Aug 5 16:54 subuid-
-rw-r--r-- 1 root root 4343 Jun 25 12:42 sudo.conf
-r--r----- 1 root root 1800 Jan 29 2024 sudoers
drwxr-xr-x 2 root root 4096 Aug 5 17:02 sudoers.d
-rw-r--r-- 1 root root 9804 Jun 25 12:42 sudo_logsrvd.conf
drwxr-xr-x 2 root root 4096 Aug 5 17:14 supercat
-rw-r--r-- 1 root root 2209 Mar 24 2024 sysctl.conf
drwxr-xr-x 2 root root 4096 Aug 5 17:02 sysctl.d
drwxr-xr-x 2 root root 4096 Aug 5 17:14 sysstat
drwxr-xr-x 6 root root 4096 Aug 5 16:49 systemd
drwxr-xr-x 2 root root 4096 Aug 5 17:00 terminfo
drwxr-xr-x 2 root root 4096 Sep 26 07:21 thermald
-rw-r--r-- 1 root root 8 Aug 5 17:02 timezone
drwxr-xr-x 2 root root 4096 Aug 5 17:14 tmpfiles.d
drwxr-xr-x 2 root root 4096 Aug 5 17:14 ubuntu-advantage
-rw-r--r-- 1 root root 1260 Jan 27 2023 ucf.conf
drwxr-xr-x 4 root root 4096 Aug 5 17:02 udev
drwxr-xr-x 2 root root 4096 Sep 28 08:27 udisks2
drwxr-xr-x 3 root root 4096 Aug 5 17:14 ufw
-rw-r--r-- 1 root root 208 Aug 5 16:54 .updated
drwxr-xr-x 3 root root 4096 Aug 5 17:02 update-manager
drwxr-xr-x 2 root root 4096 Aug 5 17:14 update-motd.d
drwxr-xr-x 2 root root 4096 Aug 5 17:14 update-notifier
drwxr-xr-x 2 root root 4096 Sep 26 07:21 UPower
-rw-r--r-- 1 root root 1523 Aug 5 17:14 usb_modeswitch.conf
drwxr-xr-x 2 root root 4096 Aug 5 17:14 usb_modeswitch.d
lrwxrwxrwx 1 root root 16 Aug 5 17:02 vconsole.conf -> default/keyboard
drwxr-xr-x 2 root root 4096 Sep 28 06:48 vim
drwxr-xr-x 4 root root 4096 Oct 2 08:24 vmware-tools
lrwxrwxrwx 1 root root 23 Feb 26 2024 vtrgb -> /etc/alternatives/vtrgb
-rw-r--r-- 1 root root 4942 Aug 5 17:14 wgetrc
drwxr-xr-x 4 root root 4096 Aug 5 17:02 X11
-rw-r--r-- 1 root root 681 Apr 8 2024 xattr.conf
drwxr-xr-x 4 root root 4096 Aug 5 17:02 xdg
drwxr-xr-x 2 root root 4096 Aug 5 17:02 xml
-rw-r--r-- 1 root root 460 Aug 5 17:14 zsh_command_not_found

```

```

crw-rw---- 1 root dialout 4, 91 Oct 17 04:12 ttyS27
crw-rw---- 1 root dialout 4, 92 Oct 17 04:12 ttyS28
crw-rw---- 1 root dialout 4, 93 Oct 17 04:12 ttyS29
crw-rw---- 1 root dialout 4, 67 Oct 17 04:12 ttyS3
crw-rw---- 1 root dialout 4, 94 Oct 17 04:12 ttyS30
crw-rw---- 1 root dialout 4, 95 Oct 17 04:12 ttyS31
crw-rw---- 1 root dialout 4, 68 Oct 17 04:12 ttyS4
crw-rw---- 1 root dialout 4, 69 Oct 17 04:12 ttyS5
crw-rw---- 1 root dialout 4, 70 Oct 17 04:12 ttyS6
crw-rw---- 1 root dialout 4, 71 Oct 17 04:12 ttyS7
crw-rw---- 1 root dialout 4, 72 Oct 17 04:12 ttyS8
crw-rw---- 1 root dialout 4, 73 Oct 17 04:12 ttyS9
drwxr-xr-x 2 root root 60 Oct 17 04:11 ubuntu-vg
crw-rw---- 1 root kvm 10, 124 Oct 17 04:12 udmabuf
crw----- 1 root root 10, 239 Oct 17 04:12 uhid
crw----- 1 root root 10, 223 Oct 17 04:12 uinput
crw-rw-rw- 1 root root 1, 9 Oct 17 04:12 urandom
crw----- 1 root root 10, 126 Oct 17 04:12 userfaultfd
crw----- 1 root root 10, 240 Oct 17 04:12 userio
crw-rw---- 1 root tty 7, 0 Oct 17 04:12 vcs
crw-rw---- 1 root tty 7, 1 Oct 17 04:12 vcs1
crw-rw---- 1 root tty 7, 2 Oct 17 04:12 vcs2
crw-rw---- 1 root tty 7, 3 Oct 17 04:12 vcs3
crw-rw---- 1 root tty 7, 4 Oct 17 04:12 vcs4
crw-rw---- 1 root tty 7, 5 Oct 17 04:12 vcs5
crw-rw---- 1 root tty 7, 6 Oct 17 04:12 vcs6
crw-rw---- 1 root tty 7, 128 Oct 17 04:12 vcsa
crw-rw---- 1 root tty 7, 129 Oct 17 04:12 vcsa1
crw-rw---- 1 root tty 7, 130 Oct 17 04:12 vcsa2
crw-rw---- 1 root tty 7, 131 Oct 17 04:12 vcsa3
crw-rw---- 1 root tty 7, 132 Oct 17 04:12 vcsa4
crw-rw---- 1 root tty 7, 133 Oct 17 04:12 vcsa5
crw-rw---- 1 root tty 7, 134 Oct 17 04:12 vcsa6
crw-rw---- 1 root tty 7, 64 Oct 17 04:12 vcsu
crw-rw---- 1 root tty 7, 65 Oct 17 04:12 vcsu1
crw-rw---- 1 root tty 7, 66 Oct 17 04:12 vcsu2
crw-rw---- 1 root tty 7, 67 Oct 17 04:12 vcsu3
crw-rw---- 1 root tty 7, 68 Oct 17 04:12 vcsu4
crw-rw---- 1 root tty 7, 69 Oct 17 04:12 vcsu5
crw-rw---- 1 root tty 7, 70 Oct 17 04:12 vcsu6
drwxr-xr-x 2 root root 60 Oct 17 04:12 vfio
crw----- 1 root root 10, 127 Oct 17 04:12 vga_arbiter
crw----- 1 root root 10, 137 Oct 17 04:12 vhci
crw-rw---- 1 root kvm 10, 238 Oct 17 04:12 vhost-net
crw-rw---- 1 root kvm 10, 241 Oct 17 04:12 vhost-vsock
crw----- 1 root root 10, 122 Oct 17 04:12 vmci
crw-rw-rw- 1 root root 10, 121 Oct 17 04:12 vsock
crw-rw-rw- 1 root root 1, 5 Oct 17 04:12 zero
crw----- 1 root root 10, 249 Oct 17 04:12 zfs
manahil@ubuntu:~$ _

```

```
manahil@ubuntu:~$ ls -la /var
total 56
drwxr-xr-x 13 root root 4096 Sep 26 07:25 .
drwxr-xr-x 23 root root 4096 Sep 26 07:21 ..
drwxr-xr-x  2 root root 4096 Oct  3 02:56 backups
drwxr-xr-x 16 root root 4096 Sep 26 08:26 cache
drwxrwsrwt  2 root root 4096 Aug  5 17:02 crash
drwxr-xr-x 45 root root 4096 Sep 26 08:26 lib
drwxrwsr-x  2 root staff 4096 Apr 22 2024 local
lrwxrwxrwx  1 root root    9 Aug  5 16:54 lock -> /run/lock
drwxrwxr-x 10 root syslog 4096 Oct 17 04:12 log
drwxrwsr-x  2 root mail 4096 Aug  5 16:54 mail
drwxr-xr-x  2 root root 4096 Aug  5 16:54 opt
lrwxrwxrwx  1 root root    4 Aug  5 16:54 run -> /run
drwxr-xr-x  2 root root 4096 May 21 15:46 snap
drwxr-xr-x  4 root root 4096 Aug  5 17:14 spool
drwxrwsrwt  9 root root 4096 Oct 17 04:25 tmp
-rw-r--r--  1 root root 208 Aug  5 16:54 .updated
manahil@ubuntu:~$
```

```
manahil@ubuntu:~$ ls -la /tmp
total 60
drwxrwxrwt 15 root root 4096 Oct 17 04:25 .
drwxr-xr-x 23 root root 4096 Sep 26 07:21 ..
drwxrwxrwt  2 root root 4096 Oct 17 04:12 .font-unix
drwxrwxrwt  2 root root 4096 Oct 17 04:12 .ICE-unix
drwx----- 2 root root 4096 Oct 17 04:12 snap-private-tmp
drwx----- 3 root root 4096 Oct 17 04:12 systemd-private-c7598c38a7c7438db2abb9803b8c65d2-fwupd.service-sGnIMw
drwx----- 3 root root 4096 Oct 17 04:12 systemd-private-c7598c38a7c7438db2abb9803b8c65d2-ModemManager.service-MUH
drwx----- 3 root root 4096 Oct 17 04:12 systemd-private-c7598c38a7c7438db2abb9803b8c65d2-polkit.service-HueF6u
drwx----- 3 root root 4096 Oct 17 04:12 systemd-private-c7598c38a7c7438db2abb9803b8c65d2-systemd-logind.service-S
drwx----- 3 root root 4096 Oct 17 04:12 systemd-private-c7598c38a7c7438db2abb9803b8c65d2-systemd-resolved.service
drwx----- 3 root root 4096 Oct 17 04:12 systemd-private-c7598c38a7c7438db2abb9803b8c65d2-systemd-timesyncd.servic
drwx----- 3 root root 4096 Oct 17 04:25 systemd-private-c7598c38a7c7438db2abb9803b8c65d2-upower.service-6e7HkX
drwx----- 2 root root 4096 Oct 17 04:12 vmware-root_751-4290559920
drwxrwxrwt  2 root root 4096 Oct 17 04:12 .X11-unix
drwxrwxrwt  2 root root 4096 Oct 17 04:12 .XIM-unix
manahil@ubuntu:~$
```

5. List your home directory and show hidden (dot) files:

```
manahil@ubuntu:~$ ls -la ~
total 40
drwxr-x--- 6 manahil manahil 4096 Oct 17 04:22 .
drwxr-xr-x 3 root root 4096 Sep 26 07:25 ..
-rw-r--r-- 1 manahil manahil 220 Mar 31 2024 .bash_logout
-rw-r--r-- 1 manahil manahil 3771 Mar 31 2024 .bashrc
drwx----- 2 manahil manahil 4096 Sep 26 07:26 .cache
drwxrwxr-x 7 manahil manahil 4096 Sep 27 12:20 .git
drwxrwxr-x 3 manahil manahil 4096 Sep 27 02:53 .local
-rw-r--r-- 1 manahil manahil 807 Mar 31 2024 .profile
-rw-rw-r-- 1 manahil manahil 20 Sep 27 12:19 README.md
drwx----- 2 manahil manahil 4096 Sep 28 14:18 .ssh
-rw-r--r-- 1 manahil manahil 0 Oct 17 04:22 .sudo_as_admin_successful
manahil@ubuntu:~$
```

5. Write a short paragraph (3–5 sentences) that explains the difference between /bin, /usr/bin and /usr/local/bin. Open your editor:

```
GNU nano 7.2 /home/manahil/answers.md
the /bin directory contains essential command binaries needed for system booting and basic operations, such as ls and cp.
the /usr/bin directory holds most of the user commands and application binaries that are not required for booting but are in normal multi-user operation.
the /usr/local/bin directory is for programs installed manually by the system administrator and not managed by the system's package manager.
```

Task 4

1. Create a workspace and navigate:

```
manahil@ubuntu:~$ mkdir -p ~/lab4/workspace/python_project
manahil@ubuntu:~$ _
```

Pwd

```
manahil@ubuntu:~/lab4/workspace/python_project$ _
```

```
manahil@ubuntu:~/lab4/workspace/python_project$ pwd
/home/manahil/lab4/workspace/python_project
manahil@ubuntu:~/lab4/workspace/python_project$
```

2. Create files using an editor (open each editor session and save a screenshot showing content):

nano README.md

```
GNU nano 7.2
Lab 4 README
```

nano main.py

```
GNU nano 7.2
print("hello lab4")_
```

nano .env

```
GNU nano 7.2
ENV=lab4
```

3. List files and capture:

ls -la

```
manahil@ubuntu:~/lab4/workspace/python_project$ ls -la
total 20
drwxrwxr-x 2 manahil manahil 4096 Oct 17 05:11 .
drwxrwxr-x 3 manahil manahil 4096 Oct 17 04:53 ..
-rw-rw-r-- 1 manahil manahil   9 Oct 17 05:11 .env
-rw-rw-r-- 1 manahil manahil  20 Oct 17 05:09 main.py
-rw-rw-r-- 1 manahil manahil  13 Oct 17 05:07 README.md
manahil@ubuntu:~/lab4/workspace/python_project$
```

Copy, move and remove:

```
cp: cannot stat 'README': No such file or directory
manahil@ubuntu:~/lab4/workspace/python_project$ cp README.md README.copy.md
manahil@ubuntu:~/lab4/workspace/python_project$
```

```
manahil@ubuntu:~/lab4/workspace/python_project$ rm README.dev.md
manahil@ubuntu:~/lab4/workspace/python_project$ _
```

mkdir -p ~/lab4/workspace/java_app

```
manahil@ubuntu:~/lab4/workspace/python_project$ rm README.dev.md
manahil@ubuntu:~/lab4/workspace/python_project$ mkdir -p ~/lab4/workspace/java_app
manahil@ubuntu:~/lab4/workspace/python_project$ _
```

cp -r ~/lab4/workspace/python_project ~/lab4/workspace/java_app_copy

```
manahil@ubuntu:~/lab4/workspace/python_project$ mkdir -p ~/lab4/workspace/java_app
manahil@ubuntu:~/lab4/workspace/python_project$ cp -r ~/lab4/workspace/python_project ~/lab4/workspace/java_app_copy
manahil@ubuntu:~/lab4/workspace/python_project$ _
```

ls -la ~/lab4/workspace

```
manahil@ubuntu:~/lab4/workspace/python_project$ ls -la ~/lab4/workspace
total 20
drwxrwxr-x 5 manahil manahil 4096 Oct 17 05:41 .
drwxrwxr-x 3 manahil manahil 4096 Oct 17 04:53 ..
drwxrwxr-x 2 manahil manahil 4096 Oct 17 05:39 java_app
drwxrwxr-x 2 manahil manahil 4096 Oct 17 05:41 java_app_copy
drwxrwxr-x 2 manahil manahil 4096 Oct 17 05:30 python_project
manahil@ubuntu:~/lab4/workspace/python_project$ _
```

5. Use command history and tab completion:

History


```

manahil@ubuntu:~/lab4/workspace/python_project$ history
 1 ip a
 2 sudo apt update
 3 sudo apt install openssh-server -y
 4 sudo system enable ssh
 5 clear
 6 ip a
 7*
 8 pwd
 9 ls -la /
10 cat /etc/os-release
11 ls -la /bin
12 ls -la /sbin
13 ls -la /opt
14 ls -la /usr
15 ls -la /etc
16 ls -la /dev
17 ls -la /dev
18 ls -la /var
19 ls -la /tmp
20 ls -la ~
21 ls -la ~
22 nano ~/answers.md
23 clear
24 mkdir -p ~/lab4/workspace/python_project
25 cd ~/lab4/workspace/python_project
26 clear
27 pwd
28 nano README.md
29 nano main.py
30 nano_env.png
31 nano .env
32 ls -la
33 cp README README.copy.md
34 cp README.md README.copy.md
35 mv README.copy.md README.dev.md
36 rm README.dev.md
37 mkdir -p ~/lab4/workspace/java_app
38 cp -r ~/lab4/workspace/python_project ~/lab4/workspace/java_app_copy
39 ls -la ~/lab4/workspace
40 history

```

- Demonstrate tab completion (type partial name and press Tab) and capture that action as tab_completion.png.

```

manahil@ubuntu:~/lab4/workspace/python_project$ cd ~/lab4/workspace/python_project/
manahil@ubuntu:~/lab4/workspace/python_project$ _

```

Task 5

1. Kernel and OS:

```

manahil@ubuntu:~/lab4/workspace/python_project$ cd ~/lab4/workspace/python_project/
manahil@ubuntu:~/lab4/workspace/python_project$ uname -a
Linux ubuntu 6.8.0-85-generic #85-Ubuntu SMP PREEMPT_DYNAMIC Thu Sep 18 15:26:59 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
manahil@ubuntu:~/lab4/workspace/python_project$ _

```

2. CPU (ensure model name visible):

```

core id      : 0
cpu cores    : 1
apicid       : 0
initial apicid : 0
fpu          : yes
fpu_exception : yes
cpuid level  : 22
wp           : yes
flags        : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss
  arch_perfmon nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse
c rdrand hypervisor lahfh_lm abm 3dnowprefetch pti ssbd ibrs ibpb stibp fsgsbase tsc_adjust bmi1 avx2 smep bmi2 invpcid
c xgetbv1 xsaves arat md_clear flush_l1d arch_capabilities
bugs         : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs srbds mmio_stale_data retbleed g
bogomips     : 3600.01
clflush size : 64
cache_alignm : 64
address sizes : 45 bits physical, 48 bits virtual
power management:

processor     : 1
vendor_id     : GenuineIntel
cpu family    : 6
model         : 142
model name    : Intel(R) Core(TM) i5-8265U CPU @ 1.60GHz
stepping      : 12
microcode     : 0xffffffff
cpu MHz       : 1800.006
cache size    : 6144 KB
physical id   : 2
siblings      : 1
core id       : 0
cpu cores     : 1
apicid        : 2
initial apicid : 2
fpu           : yes
fpu_exception : yes
cpuid level   : 22
wp            : yes
flags         : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss
  arch_perfmon nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse
c rdrand hypervisor lahfh_lm abm 3dnowprefetch pti ssbd ibrs ibpb stibp fsgsbase tsc_adjust bmi1 avx2 smep bmi2 invpcid
c xgetbv1 xsaves arat md_clear flush_l1d arch_capabilities
bugs          : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs srbds mmio_stale_data retbleed g
bogomips      : 3600.01
clflush size   : 64
cache_alignm   : 64
address sizes   : 45 bits physical, 48 bits virtual
power management:

```

3. Memory:

```

manahil@ubuntu:~/lab4/workspace/python_project$ free -h
              total        used        free      shared  buff/cache   available
Mem:          3.8Gi        498Mi        2.9Gi        1.5Mi        626Mi        3.3Gi
Swap:         1.9Gi           0B        1.9Gi
manahil@ubuntu:~/lab4/workspace/python_project$ _

```

4. Disk:

df -h

```

manahil@ubuntu:~/lab4/workspace/python_project$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           387M  1.6M  386M   1% /run
/dev/mapper/ubun--vg-ubuntu--lv 9.8G  4.8G  4.5G  52% /
tmpfs           1.9G   0    1.9G   0% /dev/shm
tmpfs           5.0M   0    5.0M   0% /run/lock
/dev/sda2       1.8G  192M  1.5G  12% /boot
tmpfs           387M  12K  387M   1% /run/user/1000
manahil@ubuntu:~/lab4/workspace/python_project$

```

5. Os Release:

cat /etc/os-release

```
manahil@ubuntu:~/lab4/workspace/python_project$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.3 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.3 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
manahil@ubuntu:~/lab4/workspace/python_project$
```

6. Processes (show top lines of ps output):

```
systemd+ 655 0.0 0.3 21588 12800 ? Ss 04:17 0:01 /usr/lib/systemd/systemd-resolved
systemd+ 658 0.0 0.1 164756 7680 ? Ssl 04:17 0:00 /usr/lib/systemd/systemd-timesyncd
root 716 0.0 0.0 0 0 ? I< 04:17 0:00 [kworker/R-cfg80]
root 750 0.0 0.3 53464 12032 ? Ss 04:17 0:00 /usr/bin/VGAAuthService
root 751 0.7 0.2 315860 9344 ? Ssl 04:17 0:48 /usr/bin/vmtoolsd
root 767 0.0 0.0 0 0 ? S 04:17 0:00 [irq/57-vmw_vmci]
root 778 0.0 0.0 0 0 ? S 04:17 0:00 [irq/58-vmw_vmci]
message+ 812 0.0 0.1 9804 5376 ? Ss 04:17 0:00 @dbus-daemon --system --address=systemd: --no
root 820 0.0 0.0 0 0 ? S 04:17 0:00 [irq/16-vmwgfx]
root 821 0.0 0.0 0 0 ? I< 04:17 0:00 [kworker/R-ttm]
polkitd 823 0.0 0.1 308164 7808 ? Ssl 04:17 0:00 /usr/lib/polkit-1/polkitd --no-debug
root 835 0.0 0.2 18124 8832 ? Ss 04:17 0:00 /usr/lib/systemd/systemd-logind
root 836 0.0 0.3 468952 13440 ? Ssl 04:17 0:00 /usr/libexec/udisks2/udisksd
root 856 0.0 0.0 0 0 ? I< 04:17 0:00 [kworker/u259:2-hci0]
root 858 0.0 0.0 0 0 ? I 04:17 0:00 [kworker/u256:2-ext4-rsv-conversion]
root 890 0.0 0.0 6824 2688 ? Ss 04:17 0:00 /usr/sbin/cron -f -P
root 915 0.0 0.5 109692 23040 ? Ssl 04:17 0:00 /usr/bin/python3 /usr/share/unattended-upgrad
root 947 0.0 0.3 392088 12800 ? Ssl 04:17 0:00 /usr/sbin/ModemManager
root 1048 0.0 0.1 6948 4608 tty1 Ss 04:17 0:00 /bin/login -p --
syslog 1142 0.0 0.1 222508 5888 ? Ssl 04:17 0:00 /usr/sbin/rsyslogd -n -iNONE
root 1424 0.0 0.0 0 0 ? S 04:18 0:00 [psimon]
manahil 1426 0.0 0.2 20108 11136 ? Ss 04:18 0:00 /usr/lib/systemd/systemd --user
manahil 1427 0.0 0.0 21148 3516 ? S 04:18 0:00 (sd-pam)
manahil 1436 0.0 0.1 8788 5504 tty1 S 04:18 0:00 -bash
root 1483 0.0 0.0 0 0 ? I< 04:18 0:00 [kworker/R-tls-s]
root 1841 0.0 1.0 477764 41472 ? Ssl 04:31 0:01 /usr/libexec/fwupd/fwupd
root 1848 0.0 0.2 314044 8960 ? Ssl 04:31 0:00 /usr/libexec/upowerd
root 1869 0.0 0.1 12020 7808 ? Ss 04:34 0:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-10
root 1870 0.0 0.2 14960 10624 ? Ss 04:34 0:00 sshd: manahil [priv]
manahil 1927 0.0 0.1 15120 6848 ? S 04:34 0:00 sshd: manahil@pts/0
manahil 1928 0.0 0.1 8648 5504 pts/0 Ss+ 04:34 0:00 -bash
root 1938 0.0 0.0 0 0 ? I< 04:34 0:00 [kworker/0:2H]
root 1952 0.0 0.0 0 0 ? I 04:40 0:01 [kworker/u258:2-events_power_efficient]
root 2167 0.0 0.0 0 0 ? I 05:27 0:00 [kworker/u258:0-events_unbound]
root 2169 0.5 0.0 0 0 ? I 05:28 0:11 [kworker/0:2-events]
root 2235 0.0 0.0 0 0 ? I 05:35 0:00 [kworker/1:1-events]
root 2618 0.0 0.0 0 0 ? I 05:41 0:00 [kworker/u257:0-events_power_efficient]
root 2626 0.6 0.0 0 0 ? I 05:46 0:08 [kworker/0:3-events]
root 2627 0.0 0.0 0 0 ? I 05:46 0:00 [kworker/u257:3-events_power_efficient]
root 2638 0.0 0.0 0 0 ? I 05:55 0:00 [kworker/u258:3-events_unbound]
root 2639 0.0 0.0 0 0 ? I 05:56 0:00 [kworker/0:0-cgroup_destroy]
root 2640 0.0 0.0 0 0 ? I 05:56 0:00 [kworker/u257:1-events_power_efficient]
root 2641 0.1 0.0 0 0 ? I 05:56 0:00 [kworker/1:2-events]
root 2644 0.0 0.0 0 0 ? I< 05:56 0:00 [kworker/0:1H-kblockd]
root 2652 0.0 0.0 0 0 ? I< 06:02 0:00 [kworker/1:0H-kblockd]
root 2653 0.0 0.0 0 0 ? I 06:02 0:00 [kworker/1:0]
root 2654 0.0 0.0 0 0 ? I 06:03 0:00 [kworker/u257:2-events_power_efficient]
root 2664 0.0 0.0 0 0 ? I 06:05 0:00 [kworker/u258:1-events_power_efficient]
manahil 2669 266 0.1 10884 4480 tty1 R+ 06:07 0:00 ps aux
manahil@ubuntu:~/lab4/workspace/python_project$
```

Task 6

1. Create a new user named lab4user:

```

manahil@ubuntu:~/lab4/workspace/python_project$ sudo adduser lab4
[sudo] password for manahil:
info: Adding user `lab4' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `lab4' (1001) ...
info: Adding new user `lab4' (1001) with group `lab4 (1001)' ...
info: Creating home directory `/home/lab4' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for lab4
Enter the new value, or press ENTER for the default
    Full Name []: manahilhabib
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `lab4' to supplemental / extra groups `users' ...
info: Adding user `lab4' to group `users' ...
manahil@ubuntu:~/lab4/workspace/python_project$

```

2. Verify the user entry:

```

info: Adding user `lab4user' to group `users' ...
manahil@ubuntu:~$ getent passwd lab4user
lab4user:x:1002:1002::,/home/lab4user:/bin/bash
manahil@ubuntu:~$ _

```

3. Switch to the new user to verify login:

```

lab4user:x:1002:1002::,/home/lab4user:/bin/bash
manahil@ubuntu:~$ su - lab4user
Password:
lab4user@ubuntu:~$

```

4. From the new user you may attempt a sudo command to show that sudo is not available for this account (expected failure), e.g.:

```

lab4user@ubuntu:~$ sudo whoami
[sudo] password for lab4user:
lab4user is not in the sudoers file.
lab4user@ubuntu:~$ _

```

5. Return to the original user:

Exit

```

manahil@ubuntu:~$ _

```

6. (Optional) Remove the test user when finished:

```
sudo deluser --remove-home lab4user
```

```
manahil@ubuntu:~$ sudo deluser --remove-home lab4user
info: Looking for files to backup/remove ...
info: Removing files ...
info: Removing crontab ...
info: Removing user `lab4user' ...
manahil@ubuntu:~$ _
```

Task 7

Create a small demo script using an editor and run it

This task is optional — complete it for extra practice or extra credit. It is not required for passing the core lab tasks.

Steps (inside VM)

1. Open an editor to create the script:

```
#!/bin/bash
echo "Lab4 demo: current user is $(whoami)"
echo "current time: $(date)"
uptime
free -h
```

2. Make the script executable:

```
manahil@ubuntu:~$ chmod +x ~/run-demo.sh
manahil@ubuntu:~$ ~/run-demo.sh
```

3. Run the script as your regular user:

```
manahil@ubuntu:~$ ~/run-demo.sh
Lab4 demo: current user is manahil
current time: Sun Oct 19 01:18:42 AM UTC 2025
01:18:42 up 17 min, 1 user, load average: 0.00, 0.03, 0.12
      total        used        free      shared  buff/cache   available
Mem:    3.8Gi       457Mi       3.3Gi       1.5Mi       302Mi       3.3Gi
Swap:    1.9Gi          0B        1.9Gi
manahil@ubuntu:~$
```

4. Optionally run it with sudo:

```
manahil@ubuntu:~$ sudo ~/run-demo.sh
Lab4 demo: current user is root
current time: Sun Oct 19 01:20:21 AM UTC 2025
01:20:21 up 19 min, 1 user, load average: 0.00, 0.02, 0.10
      total        used        free      shared  buff/cache   available
Mem:    3.8Gi       461Mi       3.3Gi       1.5Mi       302Mi       3.3Gi
Swap:    1.9Gi          0B        1.9Gi
manahil@ubuntu:~$
```

Exam Evaluation Questions

1. Remote Access Verification (Cyber Login Check)

Scenario:

You are part of a SOC (Security Operations Center) investigating unauthorized access to a Linux server hosted on VMware. Prove you can securely connect and verify your identity.

Steps:

1. Connect to the Ubuntu VM remotely from your host terminal.

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.58.139' (ED25519) to the list of known hosts.
manahil@192.168.58.139's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-85-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Oct 17 04:29:03 AM UTC 2025

System load:  0.0               Processes:            217
Usage of /:   49.0% of 9.75GB   Users logged in:     1
Memory usage: 8%               IPv4 address for ens33: 192.168.58.139
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

81 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings
```

2. Verify your current user and home directory path.

```
manahil@ubuntu:~$ whoami
manahil
manahil@ubuntu:~$ echo $HOME
/home/manahil
manahil@ubuntu:~$ pwd
/home/manahil
manahil@ubuntu:~$ _
```

3. Confirm you are connected to the correct host machine.

```
manahil@ubuntu:~$ hostname
ubuntu
manahil@ubuntu:~$ uname -n
ubuntu
manahil@ubuntu:~$ _
```

2. Filesystem Inspection for Forensic Evidence

Scenario:

The incident response team suspects malicious files in system directories. You must explore the filesystem to locate and document the system's structure.

Steps:

1. Display the contents of the root directory.

```
manahil@ubuntu:~$ ls -l /
total 1994836
lrwxrwxrwx    1 root root          7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x    2 root root     4096 Feb 26  2024 bin.usr-is-merged
drwxr-xr-x    4 root root     4096 Oct  2  08:25 boot
dr-xr-xr-x    2 root root     4096 Aug  5  23:53 cdrom
drwxr-xr-x   20 root root    4120 Oct 19  01:01 dev
drwxr-xr-x  108 root root     4096 Oct 19  01:06 etc
drwxr-xr-x    4 root root     4096 Oct 19  01:06 home
lrwxrwxrwx    1 root root          7 Apr 22  2024 lib -> usr/lib
lrwxrwxrwx    1 root root          9 Apr 22  2024 lib64 -> usr/lib64
drwxr-xr-x    2 root root     4096 Feb 26  2024 lib.usr-is-merged
drwx-----   2 root root    16384 Sep 26 12:18 lost+found
drwxr-xr-x    2 root root     4096 Aug  5  16:54 media
drwxr-xr-x    2 root root     4096 Aug  5  16:54 mnt
drwxr-xr-x    2 root root     4096 Aug  5  16:54 opt
dr-xr-xr-x  273 root root          0 Oct 19  01:01 proc
drwx-----   3 root root     4096 Aug  5  17:02 root
drwxr-xr-x   27 root root      820 Oct 19  01:02 run
lrwxrwxrwx    1 root root          8 Apr 22  2024/sbin -> usr/sbin
drwxr-xr-x    2 root root     4096 Dec 11  2024/sbin.usr-is-merged
drwxr-xr-x    2 root root     4096 Sep 26  07:25 snap
drwxr-xr-x    2 root root     4096 Aug  5  16:54 srv
-rw-----   1 root root 2042626048 Sep 26  07:21 swap.img
dr-xr-xr-x   13 root root          0 Oct 19  01:01 sys
drwxrwxrwt   13 root root     4096 Oct 19  01:01 tmp
drwxr-xr-x   12 root root     4096 Aug  5  16:54 usr
drwxr-xr-x   13 root root     4096 Sep 26  07:25 var
manahil@ubuntu:~$ _
```

2. Display the OS version and release information.

```
manahil@ubuntu:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.3 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.3 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
manahil@ubuntu:~$ _
```

3. Explore and record directory listings for /bin, /sbin, /usr, /opt, /etc, /dev, /var, and /tmp.

```

chmem          genl          lvmdump        poweroff       tcpplife.bt    xfs_admin
chpasswd       getcap         lvmpollld     ppcchcalls-bpfcc  tcppretrans-bpfcc  xfs_bmap
chroot         gethostlatency-bpfcc  lvmsadc       profile-bpfcc    tcpretrans.bt    xfs_copy
cobjnew-bpfcc  gethostlatency.bt  lvmsar        pvchange        tcprrt-bpfcc      xfs_db
compactnoop-bpfcc  getpcaps       lvreduce      pvcreate        tcpstates-bpfcc   xfsdist-bpfcc
cpgr           getty          lvremove      pvdisplay       tcpsubnet-bpfcc   xfsdist.bt
cpw           groupadd       lvrename      pvmove          tcpsynbl-bpfcc    xfs_estimate
cpudist-bpfcc  groupdel       lvs           pvremove        tcpsynbl.bt       xfs_freeze
cpuunclaimed-bpfcc  groupmems     lvscan        pvresize        tcpvtop-bpfcc     xfs_fsr
cpuwalk.bt     grpck          lxc           pvs             tcpvtracer-bpfcc  xfs_growfs
criticalstat-bpfcc  grpconv       lxd           pvscan          telinit           xfs_info
cron           grpconv       lxd           pvscan          thermald          xfs_io
cryptdisks_start  grpunconv     lxd           pvscan          thin_check        xfs_logprint
cryptdisks_stop  grub-bios-setup  mdadm         pvconv          thin_delta        xfs_mdrestore
cryptsetup      grub-install    mdadm         pvhistory_helper  thin_dump         xfs_metadump
ctrlaltdel     grub-macbless   mdadm         pvunconv        thin_ls           xfs_mkfile
dbslower-bpfcc  grub-mkconfig   mdmon         pythoncalls-bpfcc  thin_metadata_size  xfs_ncheck
dbstat-bpfcc    grub-mkdevicemap  memleak-bpfcc  pythonflow-bpfcc  thin_repair       xfs_quota
dcb            grub-probe      mkdosfs       pythongc-bpfcc    thin_restore      xfs_repair
dcsnoop-bpfcc  grub-reboot     mke2fs        pythonstat-bpfcc  thin_rmap         xfs_rtcp
dcsnoop.bt     grub-set-default  mkfs          rdmaucma-bpfcc    thin_trim         xfs_scrub
dcstat-bpfcc   halt            mkfs.bfs      readahead-bpfcc  threadsnoop-bpfcc  xfs_scrub_all
deadlock-bpfcc  hardirqs-bpfcc  mkfs.btrfs    readprofile      threadsnoop.bt     xfsslower-bpfcc
debugfs        hdparm          mkfs.cramfs    reboot           tipc              xfs_spaceman
delgroup       iconvconfig      mkfs.ext2     remove-shell     tplist-bpfcc      xtables_legacy-multi
deluser        insmod           mkfs.ext3     reset-trace-bpfcc  trace-bpfcc       xtables-monitor
depmod         installkernel    mkfs.ext4     resize2fs        ttynsnoop-bpfcc   xtables-nft-multi
devlink        integritycatalog  mkfs.fat      resolvconf       tune2fs           zerofree
dhcpcd         integritysetup   mkfs.minix    rmmod            ucallis           zfsdist-bpfcc
dirtop-bpfcc   invoke-rc.d      mkfs.ntfs     rmt              u-d-c-print-pci-ids  zfsdist-bpfcc
dmeventd       ip              mkfs.xfs      rmt-tar          uflow             zic
dmidecode      ip6tables       mkhomedir_helper  rsyslog          ufw               zramctl
dmsetup        ip6tables       mkhomedir_helper  rtacct           ugc
dmstats        ip6tables       mkhomedir_helper  rtcwake          umount.udisks2

```

```

/tmp:
snap-private-tmp
systemd-private-971635505b394fd2aee7e146142d0f18-ModemManager.service-1ehJ5k
systemd-private-971635505b394fd2aee7e146142d0f18-polkit.service-gUXUz2
systemd-private-971635505b394fd2aee7e146142d0f18-systemd-logind.service-J9Gv5K
systemd-private-971635505b394fd2aee7e146142d0f18-systemd-resolved.service-T0Xc3B
systemd-private-971635505b394fd2aee7e146142d0f18-systemd-timesyncd.service-DkYWyb
vmware-root_749-4282236466

/usr:
bin games include lib lib64 libexec local sbin share src

/var:
backups cache crash lib local lock log mail opt run snap spool tmp
manahil@ubuntu:~$ _

```

4. Display all hidden files in your home directory.

```

manahil@ubuntu:~$ ls -a ~
. .answers.md .bash_logout .bashrc .cache .git lab4 .local .profile README.md run-demo.sh .ssh .sudo_as_admin
manahil@ubuntu:~$ _

```

5. Create a markdown file summarizing your findings on key binary directories.

3. Evidence Handling & File Operations

Scenario:

You are creating a sandbox environment to safely analyze and handle suspicious files collected from a compromised system.

Steps:

1. Create a structured folder hierarchy under your home directory for analysis.

```

manahil@ubuntu:~$ mkdir -p ~/analysis/workspace
manahil@ubuntu:~$ cd ~/analysis/workspace/
manahil@ubuntu:~/analysis/workspace$ pwd
/home/manahil/analysis/workspace
manahil@ubuntu:~/analysis/workspace$ _

```

2. Create three text files, including one hidden file, in your workspace.


```

nanahil@ubuntu:~/analasis/workspacce$ echo "Evidence file 1" > file1.txt
nanahil@ubuntu:~/analasis/workspacce$ echo "Evidence file 2" > file2.txt
nanahil@ubuntu:~/analasis/workspacce$ echo "Hidden suspicious data" > .hideen_file.txt
nanahil@ubuntu:~/analasis/workspacce$ ls -a
.. file1.txt file2.txt .hideen_file.txt
nanahil@ubuntu:~/analasis/workspacce$ _

```

3. Create a backup copy of one file, rename it, and then delete it after verification.

```

nanahil@ubuntu:~/analasis/workspacce$ cp file1.txt file1_backup.txt
nanahil@ubuntu:~/analasis/workspacce$ mv file1_backup.txt renamed_backaup.txt
nanahil@ubuntu:~/analasis/workspacce$ rm renamed_backup.txt
rm: cannot remove 'renamed_backup.txt': No such file or directory
nanahil@ubuntu:~/analasis/workspacce$ rm renamed_backaup.txt
nanahil@ubuntu:~/analasis/workspacce$ ls
file1.txt file2.txt
nanahil@ubuntu:~/analasis/workspacce$ _

```

4. Copy the entire workspace as an evidence backup folder.

```

nanahil@ubuntu:~/analasis/workspacce$ cp -r ~/analasis/workspacce/ ~/analasis/workspace_backup
nanahil@ubuntu:~/analasis/workspacce$ ls ~/analasis/
workspace workspace_backup
nanahil@ubuntu:~/analasis/workspacce$

```

5. Display your command history to document all actions performed.

```

anahil@ubuntu:~/analysis/workspacce$ history
 1 sudo adduser lab4user
 2 getent passwd lab4user
 3 su - lab4user
 4 sudo deluser --remove-home lab4user
 5 nano ~/lab4/workspace/run/run-demo.sh
 6 nano run-demo.sh
 7 chmod +x ~/lab4/workspace/run-demo.sh
 8 chmod +x ~/lab4/run-demo.sh
 9 chmod +x ~/run-demo.sh
10 ~/run-demo.sh
11 sudo ~/run-demo.sh
12 whoami
13 echo $HOME
14 pwd
15 hostname
16 uname -n
17 ls -l /
18 cat /etc/os-release
19 ls /bin /sbin /usr /opt /etc /dev /var /tmp
20 ls -a ~
21 ls -a ~/analysis/workspace
22 mkdir -p ~/analysis/workspacce
23 cd ~/analysis/workspacce/
24 pwd
25 echo "Evidence file 1" > file1.txt
26 echo "Evidence file 2" > file2.txt
27 echo "Hidden suspicious data" > .hideen_file.txt
28 ls -a
29 cp file1.txt file1_backup.txt
30 mv file1_backup.txt renamed_backaup.txt
31 rm renamed
32 cp file1.txt file1_backup.txt
33 mv file1_backup.txt renamed_backaup.txt
34 rm renamed_backup.txt
35 rm renamed_backaup.txt
36 ls
37 cp -r ~/analysis/workspacce/ ~/analysis/workspace_backup
38 cp -r ~/analysis/workspacce/ ~/analysis/workspace_backup
39 ls ~/analysis/
40 history
anahil@ubuntu:~/analysis/workspacce$ _

```

6. Demonstrate Linux auto-completion by typing a partial command or filename.

```

anahil@ubuntu:~/analysis/workspacce$ cat fi_
anahil@ubuntu:~/analysis/workspacce$ cat file.txt

```

4. System Profiling and Process Monitoring

Scenario:

You are investigating a potential malware infection that is consuming excessive resources on the Linux VM.

Steps:

1. Display the system's OS and kernel version for the investigation report.

```
manahil@ubuntu:~/analisis/workspace$ uname -a
Linux ubuntu 6.8.0-85-generic #85-Ubuntu SMP PREEMPT_DYNAMIC Thu Sep 18 15:26:59 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
manahil@ubuntu:~/analisis/workspace$ _
```

2. Display CPU, memory, and disk usage information.

```
3 root      20   0      0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
4 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_g
5 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_p
6 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_
7 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
11 root     20   0      0      0      0 I   0.0   0.0   0:00.01 kworker/u256:0-ext4-rsv-conversion
12 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_pe
13 root     20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
14 root     20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
15 root     20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
16 root     20   0      0      0      0 S   0.0   0.0   0:00.48 ksoftirqd/0
17 root     20   0      0      0      0 I   0.0   0.0   0:00.58 rcu_preempt
18 root      rt   0      0      0      0 S   0.0   0.0   0:00.03 migration/0
19 root    -51   0      0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
20 root     20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
21 root     20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
22 root    -51   0      0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
23 root      rt   0      0      0      0 S   0.0   0.0   0:03.03 migration/1
24 root     20   0      0      0      0 S   0.0   0.0   0:00.05 ksoftirqd/1
29 root     20   0      0      0      0 S   0.0   0.0   0:00.01 kdevtmpfs
30 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-inet_
31 root     20   0      0      0      0 S   0.0   0.0   0:00.00 kauditd
33 root     20   0      0      0      0 S   0.0   0.0   0:00.00 khungtaskd
35 root     20   0      0      0      0 S   0.0   0.0   0:00.00 oom_reaper
37 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-write
38 root     20   0      0      0      0 S   0.0   0.0   0:00.24 kcompactd0
39 root     25   5      0      0      0 S   0.0   0.0   0:00.00 ksmd
42 root     39  19      0      0      0 S   0.0   0.0   0:00.00 khugepaged
43 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-kinte
44 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-kbloc
45 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-blkcg
46 root    -51   0      0      0      0 S   0.0   0.0   0:00.00 irq/9-acpi
47 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-tpm_d
48 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-ata_s
49 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-md
50 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-md_bi
51 root      0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-edac-

manahil@ubuntu:~/analisis/workspace$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.8Gi        476Mi        3.2Gi         1.5Mi         339Mi         3.3Gi
Swap:          1.9Gi           0B          1.9Gi

manahil@ubuntu:~/analisis/workspace$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            387M  1.5M  386M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv 9.8G  4.8G  4.5G  52% /
tmpfs            1.9G   0  1.9G   0% /dev/shm
tmpfs            5.0M   0  5.0M   0% /run/lock
/dev/sda2        1.8G  192M  1.5G  12% /boot
tmpfs            387M  12K  387M   1% /run/user/1000

manahil@ubuntu:~/analisis/workspace$
```

3. Display all active running processes to identify suspicious activity.

```

root      443  0.0  0.4  66760 19104 ?      S<S  01:01  0:01 /usr/lib/systemd/systemd-journald
root      467  0.0  0.0      0      0 ?      I<   01:01  0:00 [kworker/R-kmpat]
root      468  0.0  0.0      0      0 ?      I<   01:01  0:00 [kworker/R-kmpat]
root      495  0.0  0.6 354628 27392 ?      SLsl 01:01  0:01 /sbin/multithread -d -s
root      505  0.0  0.0      0      0 ?      I    01:01  0:00 [kworker/u256:1]
root      508  0.0  0.2  30316  8960 ?      Ss   01:01  0:01 /usr/lib/systemd/systemd-udev
root      521  0.0  0.0      0      0 ?      S    01:01  0:00 [psimon]
root      574  0.0  0.0      0      0 ?      I<   01:01  0:00 [kworker/1:2H-kblockd]
root      601  0.0  0.0      0      0 ?      S    01:01  0:00 [jbd2/sda2-0]
root      602  0.0  0.0      0      0 ?      I<   01:01  0:00 [kworker/R-ext4-]
systemd+  642  0.0  0.2  19008  9344 ?      Ss   01:01  0:00 /usr/lib/systemd/systemd-networkd
systemd+  659  0.0  0.3  21588 12800 ?      Ss   01:01  0:00 /usr/lib/systemd/systemd-resolved
systemd+  664  0.0  0.1  91024  7808 ?      Ssl  01:01  0:00 /usr/lib/systemd/systemd-timesyncd
root      708  0.0  0.0      0      0 ?      I<   01:01  0:00 [kworker/R-cfg80]
root      748  0.0  0.2  53464 11776 ?      Ss   01:01  0:00 /usr/bin/VGAuthService
root      749  0.5  0.2 241764  8832 ?      Ssl  01:01  0:15 /usr/bin/vmtoolsd
root      793  0.0  0.0      0      0 ?      S    01:01  0:00 [irq/57-vmw_vhci]
root      795  0.0  0.0      0      0 ?      S    01:01  0:00 [irq/58-vmw_vhci]
message+  802  0.0  0.1   9800  5376 ?      Ss   01:01  0:00 /usr/sbin/udisks2/udisksd
polkitd   820  0.0  0.2 308164  8064 ?      Ssl  01:01  0:00 /usr/lib/polkit-1/polkitd --no-debug
root      830  0.0  0.2  18140  8832 ?      Ss   01:01  0:00 /usr/lib/systemd/systemd-logind
root      831  0.0  0.3 468988 13312 ?      Ssl  01:01  0:00 /usr/libexec/udisks2/udisksd
syslog    857  0.0  0.1 222508  6016 ?      Ssl  01:01  0:00 /usr/sbin/rsyslogd -n -iNONE
root      861  0.0  0.0      0      0 ?      S    01:01  0:00 [irq/16-vmwgfx]
root      862  0.0  0.0      0      0 ?      I<   01:01  0:00 [kworker/R-ttm]
root      868  0.0  0.0   6824  2688 ?      Ss   01:01  0:00 /usr/sbin/cron -f -P
root      882  0.0  0.5 109692 23040 ?      Ssl  01:01  0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unatten
root      900  0.0  0.3 392028 12928 ?      Ssl  01:01  0:00 /usr/sbin/ModemManager
root      984  0.0  0.1   6976  4608 tty1   Ss   01:01  0:00 /bin/login -p --
root     1119  0.0  0.0      0      0 ?      I<   01:01  0:00 [kworker/u259:2-hci0]
root     1406  0.0  0.0      0      0 ?      S    01:01  0:00 [psimon]
manahil   1409  0.0  0.2 20100 11136 ?      Ss   01:01  0:00 /usr/lib/systemd/systemd --user
manahil   1410  0.0  0.0  21152  3516 ?      S    01:01  0:00 (sd-pam)
manahil   1419  0.0  0.1  10136  6272 tty1   S    01:01  0:00 -bash
root     1464  0.0  0.0      0      0 ?      I<   01:01  0:00 [kworker/R-tls-s]
root     1529  0.0  0.0   6104  1920 tty6   Ss+  01:05  0:00 /sbin/agetty -o -p -- \u --noclear - linux
root     1561  0.0  0.0      0      0 ?      I    01:11  0:00 [kworker/u257:0-events_power_efficient]
root     1606  0.0  0.0      0      0 ?      I    01:29  0:00 [kworker/u257:2-events_power_efficient]
root     1610  0.0  0.0      0      0 ?      I    01:30  0:00 [kworker/1:2-events]
root     1621  0.0  0.0      0      0 ?      I    01:35  0:00 [kworker/u257:1-events_freezable_power_]
root     1622  0.4  0.0      0      0 ?      I    01:35  0:03 [kworker/0:2-mm_percpu_wq]
root     1632  0.3  0.0      0      0 ?      I    01:36  0:02 [kworker/1:1-events]
root     1637  0.8  1.0 594912 43196 ?      Ssl  01:36  0:05 /usr/libexec/fwupd/fwupd
root     1644  0.0  0.2 314040  9088 ?      Ssl  01:36  0:00 /usr/libexec/upowerd
root     1678  0.0  0.0      0      0 ?      I    01:37  0:00 [kworker/u258:0-events_unbound]
root     1692  0.0  0.0      0      0 ?      I    01:41  0:00 [kworker/0:0-events]
root     1717  0.5  0.0      0      0 ?      I    01:46  0:00 [kworker/0:1-events]
root     1736  0.0  0.0      0      0 ?      I    01:48  0:00 [kworker/u257:3-events_freezable_power_]
manahil   1737  150  0.1 10884  4480 tty1   R+   01:49  0:00 ps aux

```

5. User Account Audit & Privilege Escalation Simulation

Scenario:

You are performing a **user activity audit** on a compromised Linux server.

The SOC suspects a newly created account (lab4user) may have been used for unauthorized access.

Your task is to simulate the account creation, perform privilege tests, and analyze authentication logs for forensic evidence.

Steps:

1. Create a new test user named lab4user.

```

manahil@ubuntu:~/analysis/workspacce$ sudo adduser lab4user
[sudo] password for manahil:
info: Adding user `lab4user' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `lab4user' (1002) ...
info: Adding new user `lab4user' (1002) with group `lab4user (1002)' ...
info: Creating home directory `/home/lab4user' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for lab4user
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `lab4user' to supplemental / extra groups `users' ...
info: Adding user `lab4user' to group `users' ...
manahil@ubuntu:~/analysis/workspacce$ _

```

2. Verify that the new user record exists in the system's user database.

```

manahil@ubuntu:~/analysis/workspacce$ getent passwd lab4user
lab4user:x:1002:1002::,/home/lab4user:/bin/bash
manahil@ubuntu:~/analysis/workspacce$

```

3. Log in as lab4user and confirm successful login.

```

manahil@ubuntu:~/analysis/workspacce$ su - lab4user
Password:
lab4user@ubuntu:~$ _

```

4. Attempt to run an administrative command as lab4user (expect permission denied).

```

manahil@ubuntu:~/analysis/workspacce$ su - lab4user
Password:
lab4user@ubuntu:~$ sudo ls /root
[sudo] password for lab4user:
lab4user is not in the sudoers file.
lab4user@ubuntu:~$

```

5. Switch back to your main analyst account.

```

manahil@ubuntu:~/analysis/workspacce$

```

6. Inspect the system authentication logs located at /var/log/auth.log to determine whether the lab4user account attempted any logins (successful or failed).

```

manahil@ubuntu:~/analisis/workspacce$ cat /var/log/auth.log | grep lab4user
2025-10-19T01:02:11.571702+00:00 ubuntu sudo: manahil : TTY=ttty1 ; PWD=/home/manahil ; USER=root ; COMMAND=/usr/sbin/adduser lab4user
2025-10-19T01:02:11.951746+00:00 ubuntu groupadd[1477]: group added to /etc/group: name=lab4user, GID=1002
2025-10-19T01:02:11.958847+00:00 ubuntu groupadd[1477]: group added to /etc/gshadow: name=lab4user
2025-10-19T01:02:11.962378+00:00 ubuntu groupadd[1477]: new group: name=lab4user, GID=1002
2025-10-19T01:02:12.030877+00:00 ubuntu useradd[1484]: new user: name=lab4user, UID=1002, GID=1002, home=/home/lab4user, shell=/bin/bash, from=/
2025-10-19T01:02:21.530753+00:00 ubuntu passwd[1497]: pam_unix(passwd:chauthtok): password changed for lab4user
2025-10-19T01:02:35.219501+00:00 ubuntu chfn[1498]: changed user 'lab4user' information
2025-10-19T01:02:38.307638+00:00 ubuntu gpasswd[1506]: members of group users set by root to lab4,lab4user
2025-10-19T01:04:17.318611+00:00 ubuntu su[1513]: (to lab4user) manahil on ttty1
2025-10-19T01:04:17.323411+00:00 ubuntu su[1513]: pam_unix(su-l:session): session opened for user lab4user(uid=1002) by manahil(uid=1000)
2025-10-19T01:04:54.995988+00:00 ubuntu sudo: lab4user : user NOT in sudoers ; TTY=ttty1 ; PWD=/home/lab4user ; USER=root ; COMMAND=/usr/bin/whoa
2025-10-19T01:05:43.684113+00:00 ubuntu su[1513]: pam_unix(su-l:session): session closed for user lab4user
2025-10-19T01:06:28.435397+00:00 ubuntu sudo: manahil : TTY=ttty1 ; PWD=/home/manahil ; USER=root ; COMMAND=/usr/sbin/deluser --remove-home lab4
2025-10-19T01:06:28.786703+00:00 ubuntu userdel[1543]: delete user 'lab4user'
2025-10-19T01:06:28.787807+00:00 ubuntu userdel[1543]: delete 'lab4user' from group 'users'
2025-10-19T01:06:28.792596+00:00 ubuntu userdel[1543]: removed group 'lab4user' owned by 'lab4user'
2025-10-19T01:06:28.792792+00:00 ubuntu userdel[1543]: removed shadow group 'lab4user' owned by 'lab4user'
2025-10-19T01:06:28.792914+00:00 ubuntu userdel[1543]: delete 'lab4user' from shadow group 'users'
2025-10-19T01:51:14.519943+00:00 ubuntu sudo: manahil : TTY=ttty1 ; PWD=/home/manahil/analisis/workspacce ; USER=root ; COMMAND=/usr/sbin/adduser
2025-10-19T01:51:14.736895+00:00 ubuntu groupadd[1844]: group added to /etc/group: name=lab4user, GID=1002
2025-10-19T01:51:14.741077+00:00 ubuntu groupadd[1844]: group added to /etc/gshadow: name=lab4user
2025-10-19T01:51:14.751346+00:00 ubuntu groupadd[1844]: new group: name=lab4user, GID=1002
2025-10-19T01:51:14.798745+00:00 ubuntu useradd[1851]: new user: name=lab4user, UID=1002, GID=1002, home=/home/lab4user, shell=/bin/bash, from=/
2025-10-19T01:51:20.670689+00:00 ubuntu passwd[1864]: pam_unix(passwd:chauthtok): password changed for lab4user
2025-10-19T01:51:23.841833+00:00 ubuntu chfn[1886]: changed user 'lab4user' information
2025-10-19T01:51:25.152771+00:00 ubuntu gpasswd[1895]: members of group users set by root to lab4,lab4user
2025-10-19T01:53:26.896964+00:00 ubuntu su[1904]: (to lab4user) manahil on ttty1
2025-10-19T01:53:26.899173+00:00 ubuntu su[1904]: pam_unix(su-l:session): session opened for user lab4user(uid=1002) by manahil(uid=1000)
2025-10-19T01:54:24.547105+00:00 ubuntu sudo: lab4user : user NOT in sudoers ; TTY=ttty1 ; PWD=/home/lab4user ; USER=root ; COMMAND=/usr/bin/ls /
2025-10-19T01:55:03.863651+00:00 ubuntu su[1904]: pam_unix(su-l:session): session closed for user lab4user
manahil@ubuntu:~/analisis/workspacce$

```

7. (Optional) Remove the lab4user account after the audit and verify deletion.

```

manahil@ubuntu:~/analisis/workspacce$ sudo deluser lab4user
info: Removing crontab ...
info: Removing user `lab4user' ...
manahil@ubuntu:~/analisis/workspacce$ sudo rm -r /home/lab4user
manahil@ubuntu:~/analisis/workspacce$

```
